Wireless Communication Recap

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FACULTY OF SCIENCE AND TECHNOLOGY

MM1
MM2
MM3
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Exam Preparation

Wireless Communication System Model

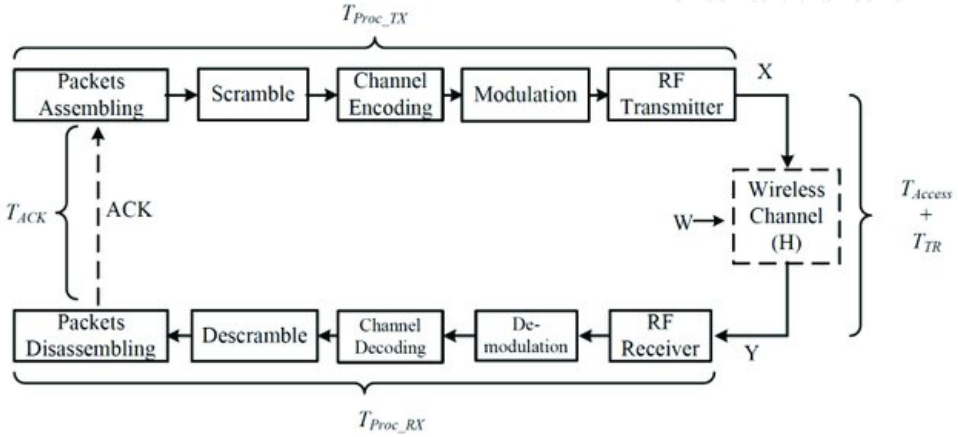

Figure: End-to-end wireless communication system: from packet generation to reception over a wireless channel.

Signals in the Time Domain

A signal is a function of time that carries information.

In the time domain, we observe:

- Amplitude
- Period / frequency
- Phase
- Waveform shape

Examples:

- Sinusoidal signals
- Square and pulse signals
- Modulated waveforms

Signals in the Frequency Domain

The frequency domain shows **how signal energy is distributed over frequency**.

In the frequency domain, we observe:

- Spectral components
- Bandwidth
- Harmonics
- Noise floor

Key ideas:

- Time and frequency domains contain the same information
- Sharp transitions in time $\rightarrow$ wide spectra
- Bandwidth is a limited and valuable resource

Fourier Transform

The Fourier Transform allows us to move between:

- Time domain $x(t)$
- Frequency domain $X(f)$

Forward and Inverse Transform

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi ft}\, df$$

Frequency Shift Property

$$x(t)\, e^{j2\pi f_0 t} \longleftrightarrow X(f - f_0)$$

This property is the basis of modulation in wireless systems.

Power of Signals

Signals can be classified by how their **energy** and **power** are distributed over time.

Power

$$P = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt$$

In wireless systems, we usually work with **power**.

Power in dBm

$$P_{\mathrm{dBm}} = 10 \log_{10} \left(\frac{P}{1\,\mathrm{mW}} \right)$$

Logarithmic units make it easier to handle large power ranges and link budgets.

Filters

Filters are used to **control which frequencies are passed or removed**.

Common filter types:

- **Low-pass filter** — passes low frequencies
- **High-pass filter** — passes high frequencies
- **Band-pass filter** — passes a selected frequency band
- **Band-stop (notch) filter** — removes a narrow band

In wireless systems, filters are used to:

- Limit signal bandwidth
- Reduce noise and interference
- Separate channels and users

Filters shape signals in the frequency domain.

Basic Filter Types

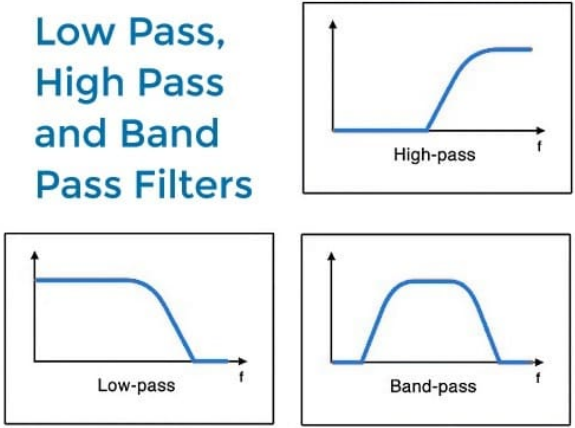

Figure: Idealised frequency responses of low-pass, high-pass, and band-pass filters.

Decibels and Power Levels

Decibels (dB) are used to express **power ratios** on a logarithmic scale.

Power ratios (dB)

$$G_{\mathrm{dB}} = 10 \log_{10}\left(\frac{P_2}{P_1}\right)$$

Absolute power (dBm)

$$P_{\mathrm{dBm}} = 10 \log_{10}\left(\frac{P}{1\,\mathrm{mW}}\right)$$

Useful reference points

- 0 dBm = 1 mW
- 10 dBm = 10 mW
- 20 dBm = 100 mW
- −10 dBm = 0.1 mW

In dB, gains and losses add instead of multiply.

Link Budget (Basic Idea)

A link budget accounts for how signal power changes from transmitter to receiver.

In its simplest form:

$$P_{\mathrm{rx}} = P_{\mathrm{tx}} + G_{\mathrm{tx}} + G_{\mathrm{rx}} - L_{\mathrm{path}} - L_{\mathrm{other}}$$

where:

- P_{tx} — transmitted power
- $G_{\mathrm{tx}}, G_{\mathrm{rx}}$ — antenna gains
- L_{path} — path loss
- L_{other} — cables, filters, mismatch, etc.

In dB, gains add and losses subtract.

What Is Noise?

In wireless communication, noise refers to **unwanted disturbances** that affect a signal.

Noise impacts:

- Signal quality
- Detection accuracy
- Achievable data rates

Common sources of noise:

- Thermal noise in electronic components
- Receiver noise (amplifiers, mixers)
- Background electromagnetic noise

Noise sets a fundamental limit on wireless communication performance.

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Noise
SNR
Noise floor
Channel Capacity
Basic Modulation
AWGN Channel model
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Exam Preparation

Additive White Gaussian Noise (AWGN)

A fundamental channel model used to analyse wireless communication systems.

The received signal is modeled as:

$$y(t) = x(t) + n(t)$$

Key characteristics

- **Additive**: noise is added to the signal
- **White**: constant power spectral density across frequency
- **Gaussian**: noise amplitude follows a normal distribution

AWGN is an idealised baseline for understanding more complex channels.

Signal-to-Noise Ratio (SNR)

The Signal-to-Noise Ratio (SNR) measures the strength of a signal relative to noise.

Definition

$$\mathrm{SNR} = \frac{P_s}{P_n}$$

SNR in decibels

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10}\left(\frac{P_s}{P_n}\right)$$

Interpretation:

- High SNR $\rightarrow$ reliable communication
- Low SNR $\rightarrow$ high error probability

Noise Power and Noise Floor

In practical systems, noise is dominated by **thermal noise**.

Noise power spectral density

$$N_0 = k_B T$$

where:

- k_B — Boltzmann's constant
- T — absolute temperature (typically 290 K)

Total noise power over bandwidth B

$$P_n = k_B T B$$

Noise floor (room temperature)

$$P_n \approx -174 + 10 \log_{10}(B) \text{ dBm}$$

Channel Capacity (Shannon Limit)

Channel capacity defines the **maximum achievable data rate** for reliable communication.

For an AWGN channel, the Shannon capacity is:

$$C = B \log_2(1 + \mathrm{SNR})$$

where:

- C — channel capacity (bits/s)
- B — bandwidth (Hz)
- SNR — signal-to-noise ratio (linear)

This is a fundamental limit — no coding or modulation can exceed it.

Power vs Bandwidth Trade-off

From Shannon's capacity formula:

$$C = B \log_2(1 + \mathrm{SNR})$$

We observe two fundamental ways to increase data rate:

- Increase **transmit power** $\rightarrow$ higher SNR
- Increase **bandwidth** $\rightarrow$ more degrees of freedom

However:

- Power is limited by regulations, hardware, and battery life
- Bandwidth is limited by spectrum availability and noise

Wireless system design is a balance between power and bandwidth.

Why Modulation Is Needed

In wireless communication, information cannot be transmitted directly as a baseband signal.

Instead, we **modulate** the information onto a carrier wave.

Modulation is used to:

- Shift signals to higher frequencies
- Enable efficient radiation with antennas
- Allow multiple users to share spectrum

At its core, modulation is frequency shifting.

Basic Modulation Concept

A carrier signal is typically a sinusoid:

$$c(t) = \cos(2\pi f_c t)$$

Information is embedded by changing one property of the carrier:

- Amplitude
- Frequency
- Phase

This creates a **bandpass signal** suitable for wireless transmission.

Basic Modulation Types

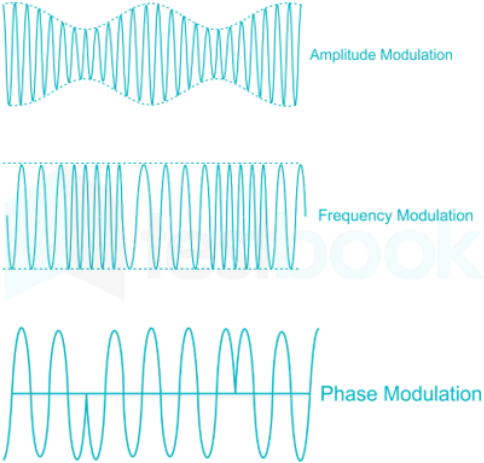
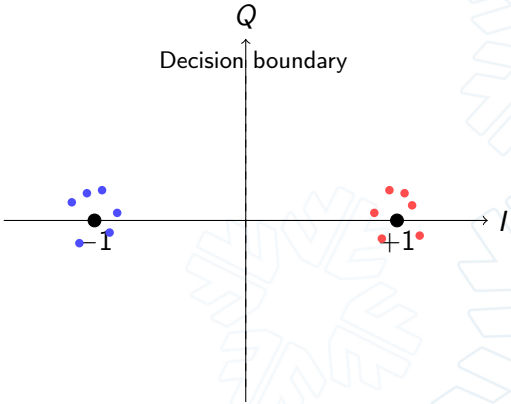


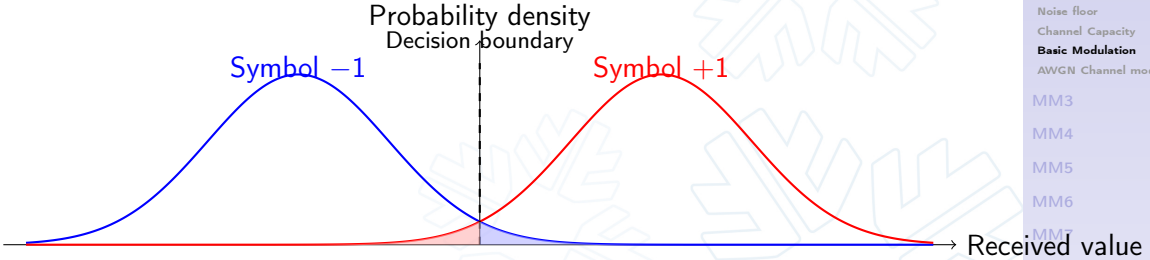
Figure: Illustration of amplitude, frequency, and phase modulation.

SNR and Bit Error Rate (BER)



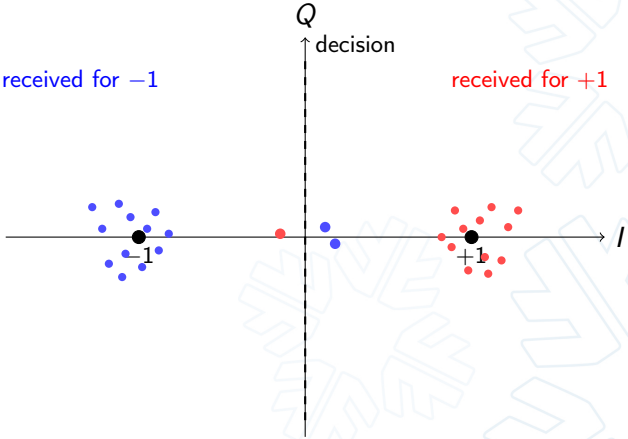
Lower SNR increases symbol spread, causing decision errors and higher BER.

Noise, Overlap, and Bit Errors



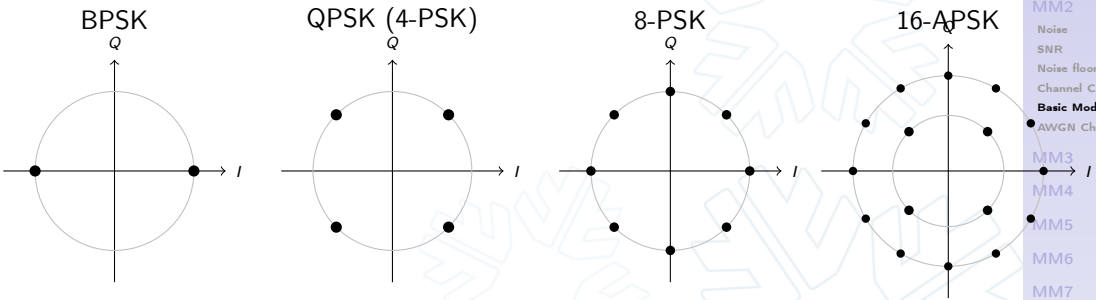
The overlapping tails represent the probability of bit errors (BER).

Constellation and SNR (BPSK Example)



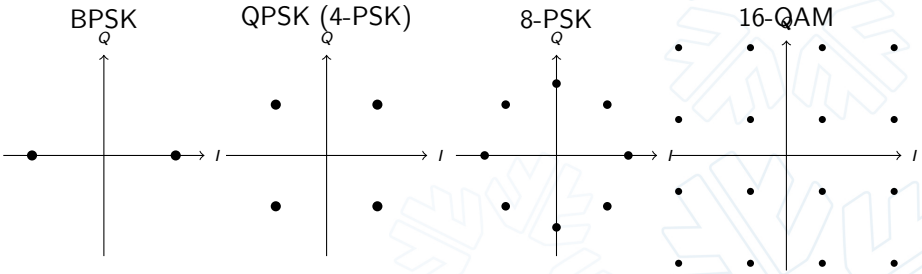
Lower SNR increases the spread, making samples cross the decision boundary ⇒ bit errors.

Constellations: BPSK, QPSK, 8-PSK, and 16-APSK



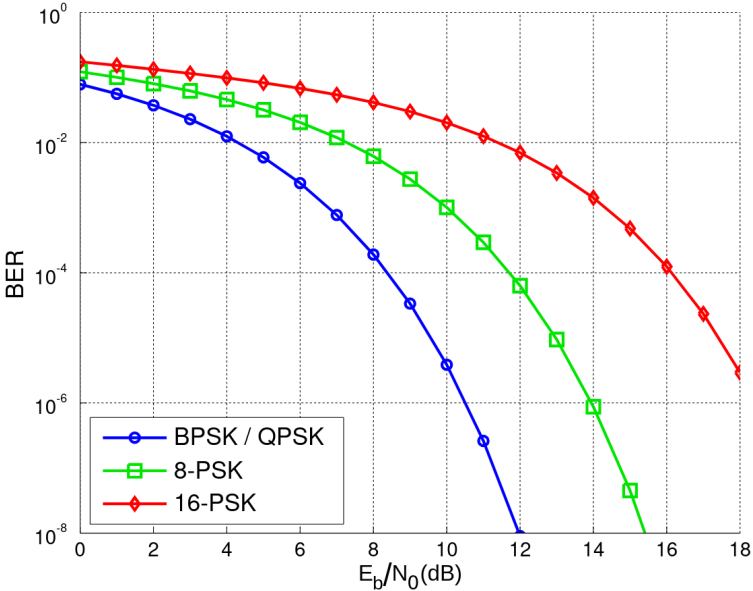
As we increase constellation size, points get closer together ⇒ higher SNR is required for the same BER.

Constellations: BPSK, QPSK, 8-PSK, and 16-QAM



Higher-order constellations pack symbols closer together, requiring higher SNR to maintain low BER.

BER vs. E_b/N_0 for PSK Modulation



Key Takeaways: Modulation and BER

From the BER curves, we observe:

- Higher-order modulation packs symbols closer together
- This reduces the minimum distance between constellation points
- Noise is therefore more likely to cause decision errors
- For the same BER:
  - BPSK / QPSK require the lowest E_b/N_0
  - 8-PSK requires higher E_b/N_0
  - 16-PSK requires significantly higher E_b/N_0

There is always a trade-off between data rate and robustness.

The AWGN Channel Model

The Additive White Gaussian Noise (AWGN) channel is a fundamental model in wireless communication.

The received signal is modeled as:

$$y(t) = x(t) + n(t)$$

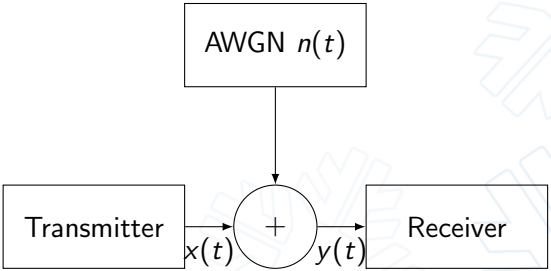
where:

- $x(t)$ is the transmitted signal
- $n(t)$ is additive noise
- $y(t)$ is the received signal

Assumptions:

- Noise is **Gaussian** with zero mean
- Noise is **white** (flat power spectral density)
- Noise is independent of the signal

AWGN Channel Model



The channel adds Gaussian noise to the transmitted signal.

Why Use the AWGN Model?

The AWGN channel is a **simplified** model, but it is extremely useful.

Why we use it

- Mathematically tractable
- Captures fundamental noise effects
- Serves as a baseline for analysis
- Widely used for performance comparison

What it does *not* include

- Fading and multipath
- Shadowing and blockage
- Interference from other users
- Mobility and Doppler effects

More realistic channel models build on AWGN.

Why Do We Need Modulation?

In wireless communication, information cannot be transmitted efficiently at baseband.

- Antennas are efficient only when their size is comparable to the wavelength
- Baseband signals would require impractically large antennas
- Multiple users must share limited spectrum resources

Solution:

- Shift the signal spectrum to a higher frequency (carrier)
- Transmit the signal as a bandpass waveform

Modulation enables efficient radiation, spectrum sharing, and multiplexing.

Baseband vs. Bandpass Signals

Baseband signal

- Contains the original information
- Centered around zero frequency
- Examples: audio, sensor data, digital symbols

Bandpass signal

- Information shifted to a carrier frequency f_c
- Occupies a frequency band around f_c
- Suitable for wireless transmission

Key idea:

$$\text{Bandpass signal} = \text{Baseband signal} \times \cos(2\pi f_c t)$$

Modulation is frequency translation from baseband to bandpass.

Bandwidth of a Bandpass Signal

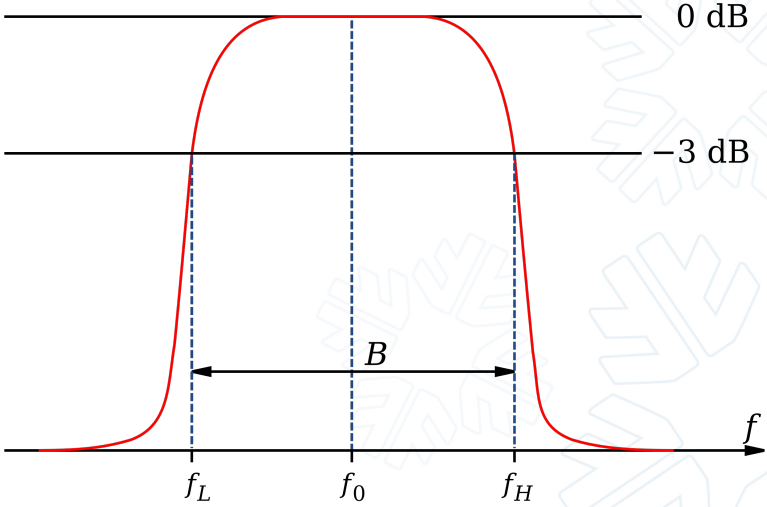


Figure: Idealized bandpass spectrum showing center frequency and occupied bandwidth.

Amplitude Modulation (AM)

Definition: Amplitude modulation is a process in which the amplitude of a carrier is varied linearly with the message signal.

Message and carrier

$$m(t): \text{ message signal, } |m(t)| \leq 1$$

$$c(t) = A_c \cos(2\pi f_c t)$$

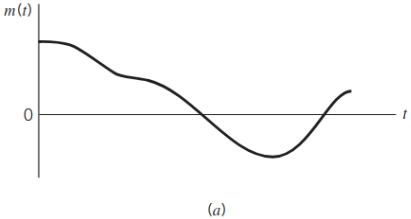
Conventional AM signal

$$s(t) = A_c \left[1 + k_a m(t)\right] \cos(2\pi f_c t)$$

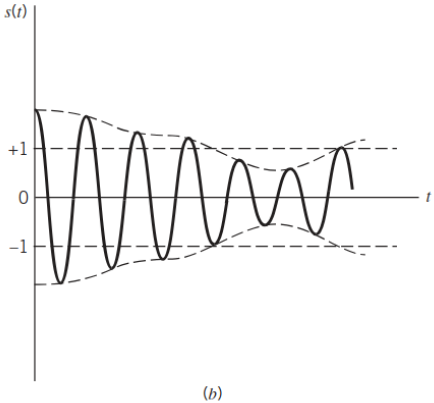
where:

- A_c is the carrier amplitude
- f_c is the carrier frequency
- k_a is the amplitude sensitivity

Amplitude Modulation: Time-Domain View

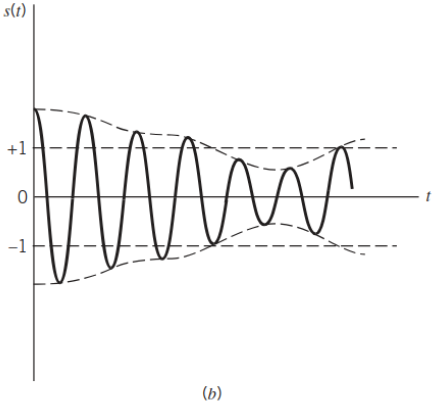


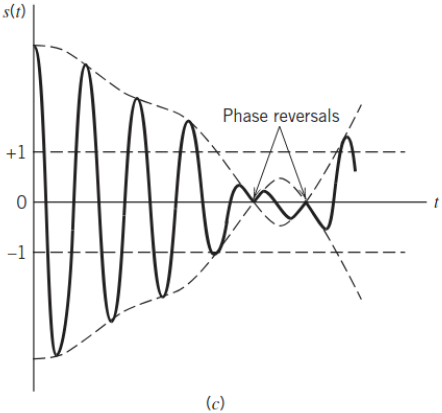
(a) Message signal $m(t)$



(b) AM signal

Overmodulation and Phase Reversal





Proper AM (no overmodulation)

$$|k_a m(t)| \leq 1$$

Overmodulated AM

Spectrum of a Conventional AM Signal

For the AM signal

$$s(t) = A_c\left[1 + k_a m(t)\right]\cos(2\pi f_c t),$$

the frequency-domain representation consists of three components.

- A **carrier** at frequency f_c
- An **upper sideband** at $f_c + f$
- A **lower sideband** at $f_c - f$

This means:

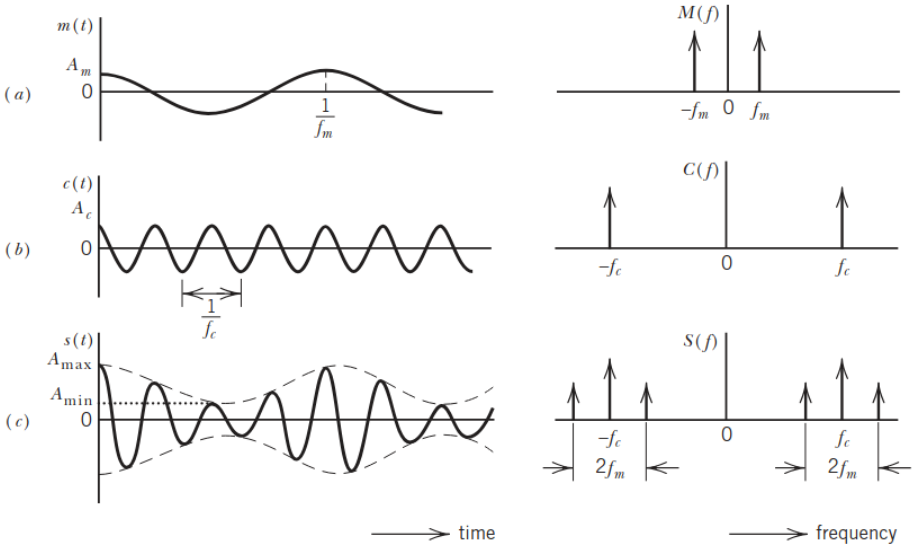
- The message bandwidth is B
- The total AM bandwidth is

$$B_{\text{AM}} = 2B$$

- Most transmitted power is wasted in the carrier

This inefficiency motivates suppressed-carrier and single-sideband modulation.

AM in Time and Frequency Domain



Double-Sideband Suppressed Carrier (DSB-SC)

Motivation

- In conventional AM, most transmitted power is wasted in the carrier
- The carrier does not contain information
- Removing the carrier improves power efficiency

DSB-SC signal model

$$s(t) = c_t m(t)$$

Key characteristics

- Same bandwidth as conventional AM ($2B$)
- No carrier component in the spectrum
- All transmitted power carries information

Main challenge

- Envelope detection no longer works
- Requires **coherent detection**
- Accurate carrier frequency and phase recovery is essential

Single-Sideband Modulation (SSB)

Motivation

- In AM and DSB-SC, the two sidebands contain redundant information
- Transmitting both sidebands wastes bandwidth
- SSB removes one sideband to improve spectral efficiency

Key properties

- Only one sideband is transmitted (USB or LSB)
- Required bandwidth is reduced to

$$B_{\text{SSB}} = B$$

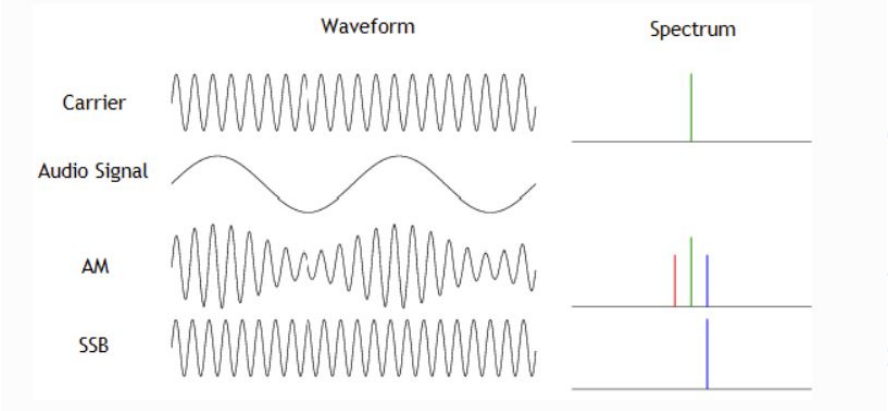
- Higher power efficiency than AM and DSB-SC

Challenges

- Generation is more complex (filtering or phase-shift methods)
- Requires coherent detection
- Sensitive to carrier frequency and phase errors

SSB is widely used where bandwidth is scarce (e.g., HF radio).

AM vs. SSB: Time and Frequency View



- AM transmits carrier and two sidebands
- Both sidebands contain the same information
- SSB transmits only one sideband and no carrier
- Bandwidth is reduced by a factor of two

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Bandpass
Amplitude Modulation
DSB-SC
SSB
Phase Modulation
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Angle Modulation: FM and PM

Key idea: Information is carried in the *angle* of the carrier, not its amplitude.

General angle-modulated signal

$$s(t) = A_c \cos\left(2\pi f_c t + \phi(t)\right)$$

- Carrier amplitude A_c is constant
- Information is encoded in the phase term $\phi(t)$

Two main types

- **Frequency Modulation (FM)** Instantaneous frequency varies with the message signal
- **Phase Modulation (PM)** Instantaneous phase varies with the message signal

Angle modulation is more robust to amplitude noise than AM.

Frequency Modulation (FM)

Definition: In frequency modulation, the instantaneous frequency of the carrier varies linearly with the message signal.

FM signal model

$$s(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right)$$

where:

- k_f is the frequency sensitivity (Hz per unit amplitude)
- $m(t)$ is the message signal

Instantaneous frequency

$$f_i(t) = f_c + k_f m(t)$$

The carrier frequency "wiggles" around f_c according to $m(t)$.

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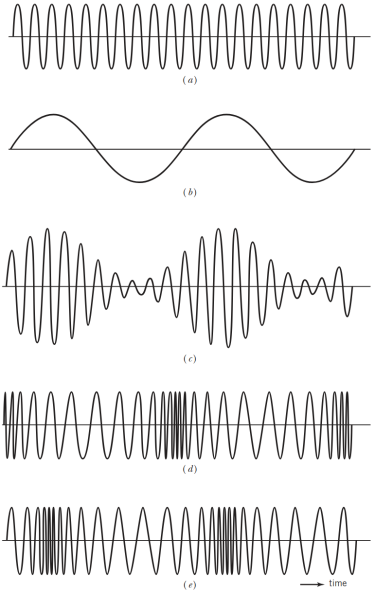

FIGURE 4.1 Illustrating AM, PM, and FM signals produced by a single tone. (*a*) Carrier wave, (*b*) sinusoidal modulating signal, (*c*) amplitude-modulated signal, (*d*) phase-modulated signal, (*e*) frequency-modulated signal.

FM and Spectrum Use

Key characteristic of FM: Improved noise robustness comes at the cost of increased bandwidth.

FM spectrum properties

- FM generates multiple sidebands around the carrier
- The number of significant sidebands increases with modulation index β
- Bandwidth grows with frequency deviation

Bandwidth (Carson's Rule)

$$B_{\text{FM}} \approx 2(\Delta f + B)$$

Implications for spectrum use

- FM occupies much more bandwidth than AM or SSB
- Wideband FM limits the number of users per frequency band
- Suitable when bandwidth is available and noise immunity is critical

FM trades spectral efficiency for robustness.

From Message Signal to Phase and Frequency Modulation

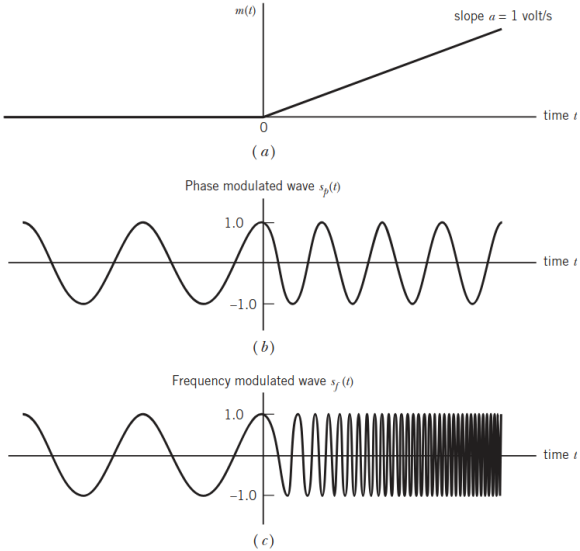

FIGURE 4.2 Starting at time $t = 0$, the figure displays (a) linearly increasing message signal $m(t)$, (b) phase-modulated wave, and (c) frequency-modulated wave.

Phase Modulation (PM)

Definition: In phase modulation, the instantaneous phase of the carrier varies linearly with the message signal.

PM signal model

$$s_p(t) = A_c \cos\big(2\pi f_c t + k_p m(t)\big)$$

where:

- A_c is the carrier amplitude
- f_c is the carrier frequency
- k_p is the phase sensitivity (rad per unit amplitude)
- $m(t)$ is the message signal

Key observations

- Carrier amplitude remains constant
- Phase changes follow the instantaneous value of $m(t)$
- PM is insensitive to amplitude noise

Relationship Between Phase Modulation and Frequency Modulation

Phase modulation and frequency modulation are closely related forms of angle modulation.

Phase modulation

$$s_p(t) = A_c \cos\big(2\pi f_c t + k_p m(t)\big)$$

Frequency modulation

$$s_f(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right)$$

- FM can be obtained by integrating the message signal before phase modulation
- PM can be obtained by differentiating the message signal before frequency modulation

PM and FM differ mainly in how the message affects the carrier phase.

From Analog Signals to Digital Information

So far, we have described **continuous-time, continuous-amplitude** signals.

In digital communication

- Information is represented by a finite set of symbols
- Symbols correspond to bits or groups of bits
- Signals change only at discrete time instants

Key shift

- From waveforms to symbol sequences
- From spectrum shape to symbol timing and detection

This transition allows reliable communication in noisy channels.

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FM
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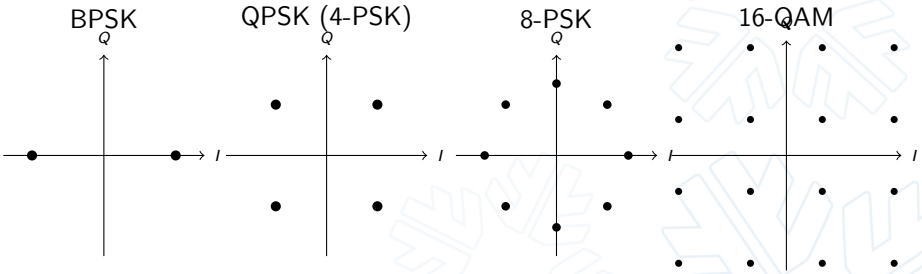
MM7
Exam Preparation

Constellations: BPSK, QPSK, 8-PSK, and 16-QAM



Higher-order constellations pack symbols closer together, requiring higher SNR to maintain low BER.

Amplitude and Phase from I/Q

Any complex baseband sample can be written as

$$s = I + jQ = Ae^{j\theta}$$

- **Amplitude (radius):**

$$A = |s| = \sqrt{I^2 + Q^2}$$

- **Phase (angle):**

$$\theta = \arg(s) = \text{atan2}(Q, I)$$

PSK: constant amplitude, varying phase. *QAM: varying amplitude and phase.*

What Is a Symbol?

- A **symbol** represents information over a finite time interval
- Symbols are not instantaneous — they last for a duration
- Different patterns (dots and dashes) represent different symbols

In wireless communication, symbols are waveforms transmitted over time.

Symbol Duration

A symbol is transmitted over a finite time interval.

Key definitions

- Symbol duration: T_s (seconds per symbol)
- Symbol rate: $R_s = \dfrac{1}{T_s}$ (symbols per second)

Relation to bit rate

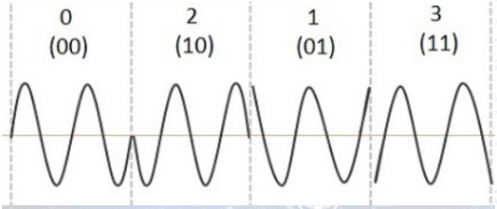
- Each symbol carries $\log_2 M$ bits
- Bit rate:

$$R_b = R_s \log_2 M$$

Why symbol duration matters

- Shorter symbols $\Rightarrow$ higher data rate
- Shorter symbols require more bandwidth
- Noise and channel distortion affect shorter symbols more strongly

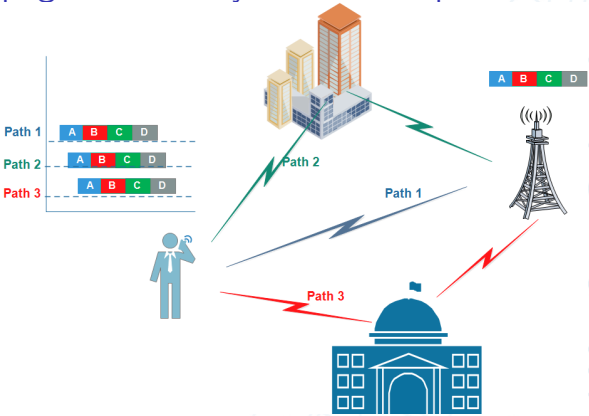
Symbols in Time



- Each symbol occupies a fixed time interval T_s
- Information is carried by the waveform over that interval
- The receiver samples once per symbol to make a decision

Symbol boundaries define when decisions are made.

Multipath Propagation and Symbol Overlap



- Signals arrive at the receiver over multiple paths
- Each path has a different delay
- Symbols can overlap in time at the receiver

This phenomenon is known as intersymbol interference (ISI).

Symbol Rate and Bandwidth (Intuition)

A faster-changing signal needs more bandwidth.

- Symbol rate: $R_s = \frac{1}{T_s}$ (baud)
- Higher R_s means faster transitions in time
- Faster transitions $\Rightarrow$ more high-frequency content

Rule of thumb

$$B \propto R_s$$

Bandwidth is the price you pay for higher symbol rate.

Spectral Efficiency

We want high data rate while using as little bandwidth as possible.

Definitions

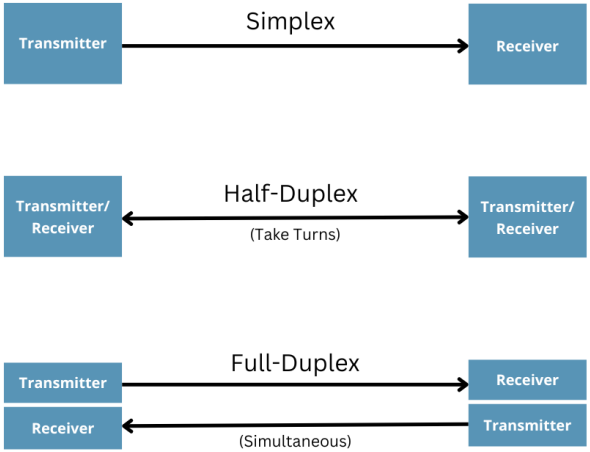
$$R_b = R_s \log_2 M \qquad \eta = \frac{R_b}{B} \text{ (bits/s/Hz)}$$

- Increasing M increases bits per symbol
- Increasing R_s increases symbols per second
- Both tend to require higher SNR and/or more bandwidth

Wireless design is a trade-off between rate, bandwidth, and robustness.

Duplexing in Wireless Communication

What is Duplexing? Duplexing defines whether a communication system can send and receive data at the same time.



Resource Allocation in Wireless Networks

Definition: Resource allocation refers to the efficient distribution of available spectrum, time, and power among multiple users in a wireless network to maximize performance and minimize interference.

Key Challenges:

- **Limited Spectrum:** Wireless spectrum is a finite resource, requiring efficient allocation.
- **Interference Management:** Multiple users accessing the network simultaneously can cause interference.
- **Quality of Service (QoS):** Ensuring reliability, low latency, and fairness across users.
- **Scalability:** Networks must support an increasing number of users and devices (e.g., IoT, 5G).

Resource Allocation Techniques

Traditional vs. Modern Approaches

- **Fixed Allocation:** Resources are pre-allocated and do not change dynamically.
  - **FDMA:** Frequency bands assigned to users.
  - **TDMA:** Time slots assigned to users.
  - **CDMA:** Unique codes assigned to users.
- **Dynamic Allocation:** Resources are adjusted based on network conditions.
  - **OFDM:** Efficiently divides spectrum into subcarriers.
  - **OFDMA:** Allows multiple users to share subcarriers dynamically.
- **Software-Defined Networking (SDN):** Enables real-time network control and optimization.

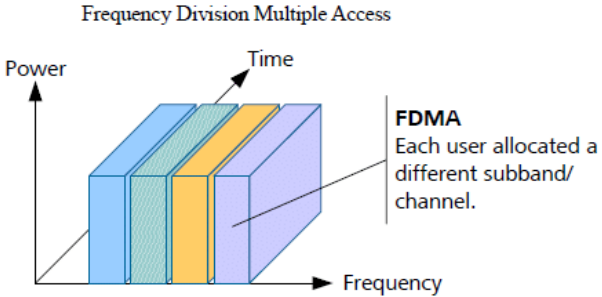
Frequency Division Multiple Access (FDMA)

Overview:

- FDMA assigns each user a unique frequency band (subchannel).
- Users transmit simultaneously but in separate frequency bands.
- No interference between users if channels are well separated.

Characteristics:

- Fixed bandwidth allocation.
- Efficient for analog signals (e.g., early mobile networks, radio broadcasting).
- Limited flexibility and inefficient for bursty traffic.



Time Division Multiple Access (TDMA)

- TDMA divides access to a frequency channel into time slots.
- Users take turns transmitting in assigned time slots.
- No interference between users as transmissions do not overlap in time.

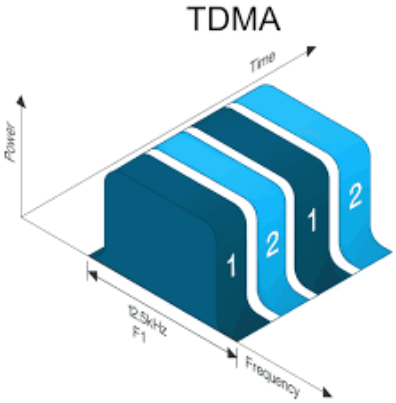


Figure: TDMA: Users transmit in different time slots on the same frequency.

Code Division Multiple Access (CDMA) - Overview

- CDMA allows multiple users to transmit on the same frequency at the same time.
- Each user is assigned a unique code to differentiate their signal.
- The receiver uses the same code to extract the intended signal while ignoring others.

Characteristics:

- More efficient than FDMA and TDMA for handling many simultaneous users.
- Requires complex signal processing to decode overlapping transmissions.
- Resistant to interference and eavesdropping due to spread spectrum techniques.

Example Use Cases:

- **3G Cellular Networks (UMTS/CDMA2000)**
- **GPS (Global Positioning System)**: Uses CDMA to allow multiple satellites to broadcast on the same frequency.

CDMA - Encoding and Decoding

Encoding: Spreading the Signal

- Each user's data signal $d(t)$ is multiplied by a unique pseudo-random code $c(t)$.
- The transmitted signal is:

$$S(t) = d(t) \cdot c(t) \tag{1}$$

Decoding: Extracting the Intended Signal

- The receiver multiplies the received signal $S(t)$ by the same code $c(t)$.

$$d_{\text{received}}(t) = S(t) \cdot c(t) = d(t) \cdot c(t) \cdot c(t) \tag{2}$$

- Since $c(t) \cdot c(t) = 1$, the original data is recovered:

$$d_{\text{received}}(t) = d(t) \tag{3}$$

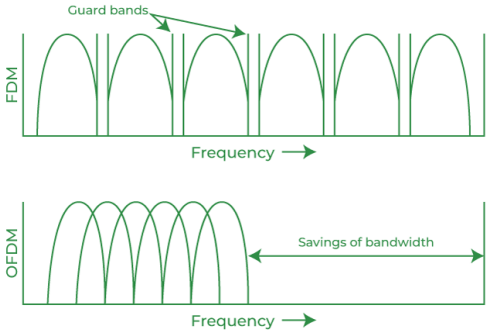
Why Does This Work?

- Other users' signals have different codes, which appear as low-level noise when decoded incorrectly.
- Only the intended code properly reconstructs the signal.

Orthogonal Frequency Division Multiplexing (OFDM)

Overview:

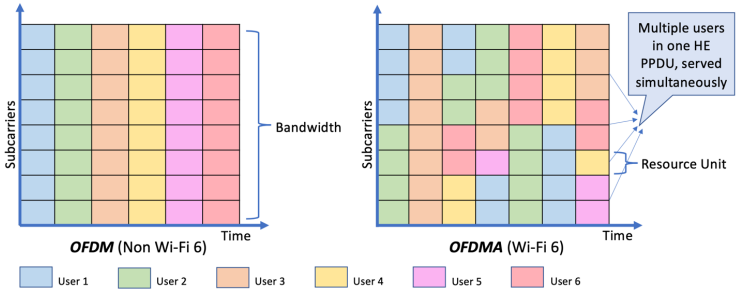
- OFDM divides the spectrum into multiple closely spaced subcarriers, eliminating the need for guard bands.
- Improves spectral efficiency and reduces interference.
- Robust against multipath fading, improving performance in wireless channels.
- Used in Wi-Fi, 4G LTE & 5G NR, Digital TV and Radio (DVB, DAB)



Orthogonal Frequency Division Multiple Access (OFDMA)

Overview:

- OFDMA extends OFDM by allowing multiple users to share subcarriers within the same time slot.
- Each user is assigned a subset of available subcarriers, optimizing resource utilization.
- Supports dynamic and flexible resource allocation per user.
- More efficient than OFDM in high-density networks.



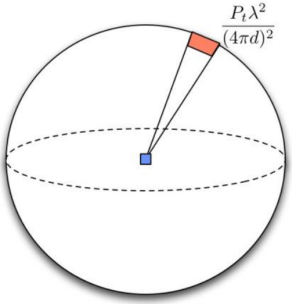
Why Media Access Control Is Hard in Wireless

Key challenge: Wireless communication takes place over a shared, unreliable medium.

- Multiple users transmit over the same channel
- Signals interfere with each other
- Channel conditions vary with time and location
- Transmitters often cannot detect collisions

To design MAC protocols, we must first understand the wireless channel.

Wireless Channel Over distance



$$P_r = G_t G_r \frac{P_t \lambda^2}{(4\pi d)^2} \tag{4}$$

where: P_t is the transmit power (dBm), G_t, G_r are the transmitter and receiver antenna gains (dBi), $\lambda = c/f$ is the wavelength (with c being the speed of light and f the frequency), d is the distance between transmitter and receiver.

Effect of Frequency

The corresponding path loss (FSPL) in dB is:

$$L_{\text{FSPL}}(dB) = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right) \tag{5}$$

$$L_{\text{FSPL}} = 20\log_{10}(d) - 20\log_{10}(f) - 147.6 \tag{6}$$

Effect of Frequency

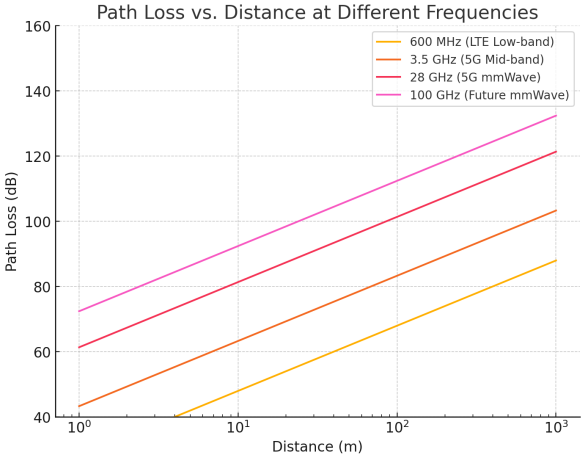
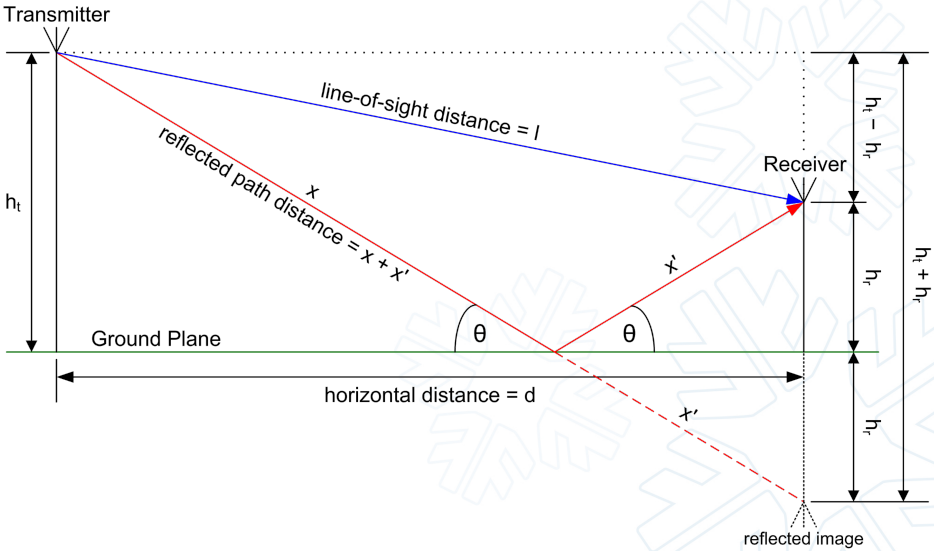


Figure: FSPL versus Distance at different frequencies


Transmitter
line-of-sight distance = l
reflected path distance = x + x'
x
x'
Receiver
h_t
$h_t - h_r$
h_r
$h_t + h_r$
Ground Plane
θ
θ
horizontal distance = d
x'
h_r
reflected image

Two-Ray Ground Reflection: Path Loss vs Distance

Two-ray received power model

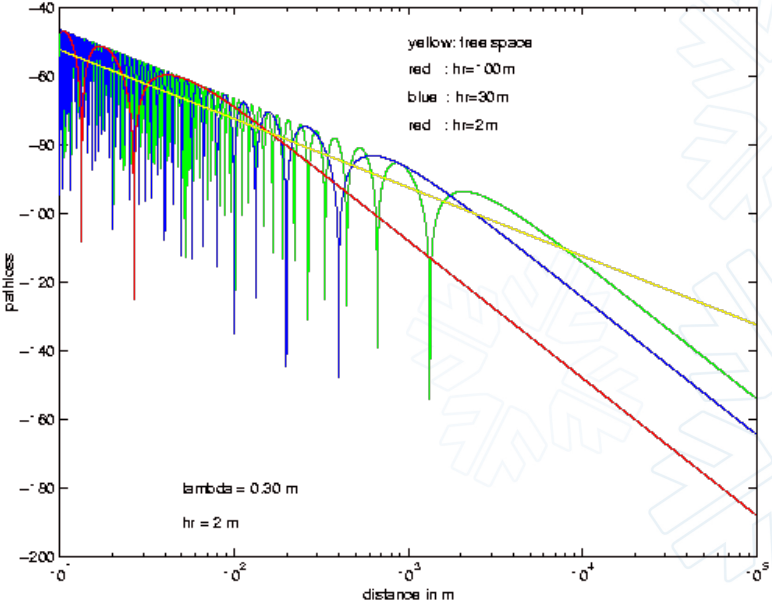
$$P_r(d) = \left(\frac{\lambda}{4\pi d}\right)^2 \cdot 4\sin^2\left(\frac{2\pi h_t h_r}{\lambda d}\right)$$

Parameters used in the plot

$$\lambda = 0.30 \text{ m}, \qquad h_r = 2 \text{ m}, \qquad h_t \in \{2,\ 30,\ 100\} \text{ m}$$

- Yellow: free-space path loss
- Colored curves: two-ray model (LOS + ground reflection)
- Oscillations caused by distance-dependent phase difference

Two-Ray Ground reflection model



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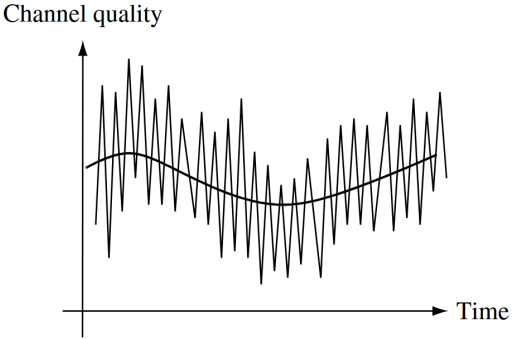
Wireless Channel Variations Over Time

Figure 2.1 Channel quality varies over multiple time-scales. At a slow scale, channel varies due to large-scale fading effects. At a fast scale, channel varies due to multipath effects.



- Wireless channel quality is **not constant over time**
- Variations occur on **multiple time scales**
  - **Slow variations:** distance, obstacles, shadowing
  - **Fast variations:** multipath interference

The Wireless Channel: Physical View

Wireless communication takes place over a **propagation channel**.

- Energy radiates from the transmitter antenna
- Waves propagate through free space and the environment
- Objects cause reflection, diffraction, and scattering

Key consequence

- The receiver observes multiple delayed and attenuated copies
- The channel is **time-varying** and **frequency "selective"**

The channel is not a single path, but a superposition of many paths.

Multipath Propagation

In realistic environments:

- Signals arrive via multiple paths
- Each path has its own attenuation and delay
- Motion causes time variation (Doppler)

Each path contributes a delayed and scaled copy of the transmitted signal.

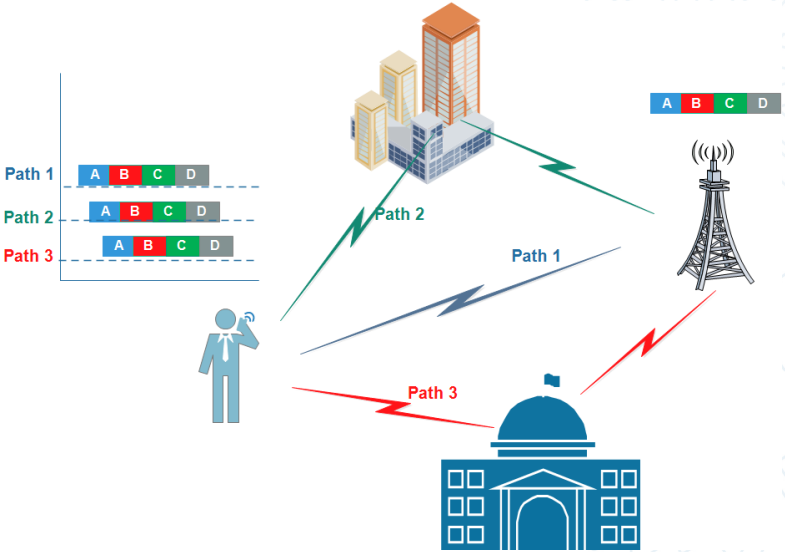
Linear Time-Varying Channel Model

Assuming linearity and narrowband signaling, the received signal can be written as:

$$y(t) = \sum_i a_i(t)\, x(t - \tau_i(t))$$

- $a_i(t)$: path attenuation (slowly varying)
- $\tau_i(t)$: propagation delay (time-varying)
- Captures multipath and Doppler effects

Linear Time-Varying Channel Model



Impulse Response of the Wireless Channel

The channel can equivalently be represented by a time-varying impulse response:

$$h(\tau, t) = \sum_i a_i(t)\,\delta(\tau - \tau_i(t))$$

The input–output relation becomes:

$$y(t) = \int h(\tau, t)\,x(t - \tau)\,d\tau$$

- Channel is linear but time-varying
- Multipath appears as discrete impulses in delay

Why This Channel Model Matters

This abstraction allows us to:

- Analyze fading, delay spread, and Doppler spread
- Define coherence time and coherence bandwidth
- Design modulation, equalization, and coding schemes

Key takeaway: All wireless channel complexity is captured by

$$\{a_i(t),\ \tau_i(t)\}$$

Power Delay Profile (PDP)

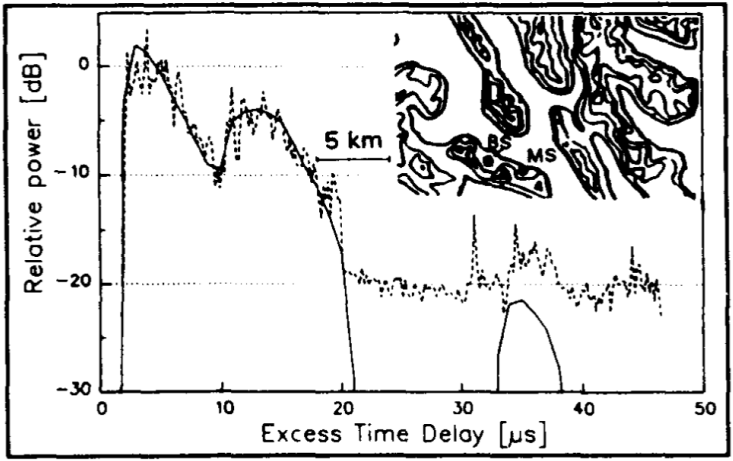


***Figure 3** : Mean PDP for an area with reflections from three mountains on two islands across a fjord. Dashed : measurement, solid : calculated.*

Delay Spread and Channel Impulse Response

The PDP is related to the channel impulse response:

$$h(\tau, t) = \sum_i a_i(t)\,\delta(\tau - \tau_i(t))$$

- $|a_i(t)|^2$: power of the i-th path
- Distribution of τ_i determines delay spread
- PDP $\approx \mathbb{E}\{|h(\tau, t)|^2\}$

Key metric:

$$T_d \text{ (delay spread)} \Rightarrow \text{ frequency selectivity}$$

Channel as a Frequency-Selective Filter

Taking the Fourier transform of the impulse response:

$$H(f,t) = \int h(\tau,t)\, e^{-j2\pi f\tau}\, d\tau$$

- Multipath delays create frequency-dependent fading
- Different frequencies experience different gains and phases
- The channel acts as a **linear time-varying filter**

Input–output relation in frequency domain:

$$Y(f) = H(f)\,X(f)$$

Delay Spread and Coherence Bandwidth

Fundamental trade-off:

$$\text{Large delay spread} \Rightarrow \text{small coherence bandwidth}$$

Rule of thumb

$$W_c \approx \frac{1}{T_d}$$

- If signal bandwidth $B \ll W_c$: flat fading
- If signal bandwidth $B > W_c$: frequency-selective fading

The Four Domains of the Wireless Channel

A wireless channel can be described in four equivalent domains.

Time–Delay	**Time–Frequency**
$h(\tau, t)$	$H(f, t)$
Delay spread T_d	Coherence bandwidth B_c
Doppler–Delay	**Frequency–Doppler**
$S(\tau, \nu)$	$H(f, \nu)$
Doppler spread B_D	Coherence time T_c

Each domain highlights different physical aspects of the channel.

Transformations Between Channel Domains

The four channel representations are related by Fourier transforms.

$$
\begin{array}{ccc}
h(\tau, t) & \xleftrightarrow{\mathcal{F}_\tau} & H(f, t) \\
\updownarrow \mathcal{F}_t & & \updownarrow \mathcal{F}_t \\
S(\tau, \nu) & \xleftrightarrow{\mathcal{F}_\tau} & H(f, \nu)
\end{array}
$$

- Delay $\leftrightarrow$ frequency: frequency selectivity
- Time $\leftrightarrow$ Doppler: time selectivity

Channel Spreads and Their Meaning

Each domain introduces a characteristic spread.

- **Delay spread** T_d: causes ISI and frequency selectivity
- **Coherence bandwidth** B_c: determines flat vs. selective fading
- **Doppler spread** B_D: caused by mobility
- **Coherence time** T_c: determines how fast the channel changes

Key dualities

$$B_c \approx \frac{1}{T_d}, \qquad T_c \approx \frac{1}{B_D}$$

These parameters drive modulation, coding, and MAC design.

Why Do We Need Cellular Systems?

- Limited spectrum
- Increasing number of users
- Limited transmit power
- Interference constraints

Key problem: How do we serve many users with limited bandwidth?

Single High-Power Transmitter

- Large coverage area
- No frequency reuse
- Capacity limited by total spectrum
- High transmit power required

Conclusion: Not scalable.

The Cellular Idea

Core Idea

Divide coverage area into smaller regions (cells) Reuse frequencies in spatially separated cells.

Advantages:

- Increased capacity
- Reduced transmit power
- Better spectral efficiency

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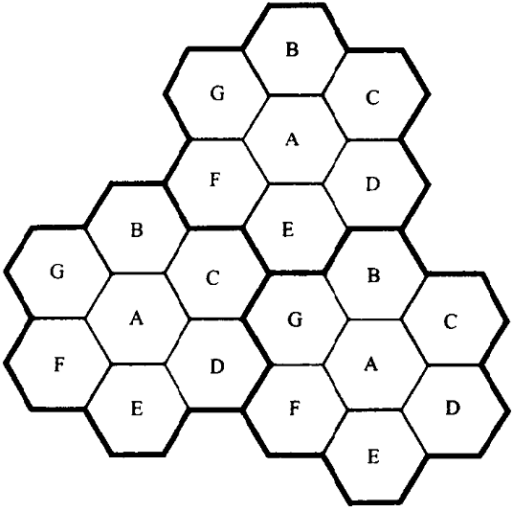

Figure 2.1
Illustration of the cellular frequency reuse concept. Cells with the same letter use the same set of frequencies. A cell cluster is outlined in bold and replicated over the coverage area. In this example, the cluster size, N, is equal to seven, and the frequency reuse factor is 1/7 since each cell contains one-seventh of the total number of available channels.

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Basic Terminology

- Cell
- Base Station (BS)
- Mobile Station (MS)
- Cluster
- Frequency reuse
- Co-channel interference
- Adjacent channel interference

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Why Hexagons?

- Circles do not tile the plane
- Squares distort distance symmetry
- Hexagons approximate circular coverage
- Equal distance to neighbors

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Cluster Size

Cluster size:

$$N = i^2 + ij + j^2$$

Examples:

$$N = 1, 3, 4, 7, 12, 19, \ldots$$

Reuse Distance

Reuse distance:

$$D = \sqrt{3N}R$$

Where:

- D = co-channel reuse distance
- R = cell radius
- N = cluster size

Geometry of Frequency Reuse

- Move i cells in one direction
- Turn 60 degrees
- Move j cells
- Arrive at co-channel cell

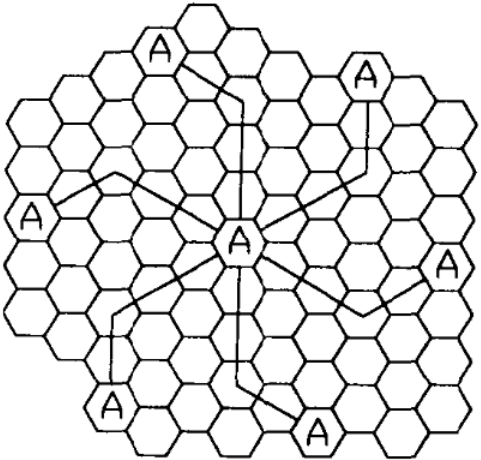

Figure 2.2
Method of locating co-channel cells in a cellular system. In this example, $N = 19$ (i.e., $i = 3, j = 2$). [Adapted from [Oet83] © IEEE].

Capacity Trade-Off

- Small $N \rightarrow$ High capacity
- Small $N \rightarrow$ High interference
- Large $N \rightarrow$ Low interference
- Large $N \rightarrow$ Lower capacity

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Why Does Large N Reduce Capacity?

Total system bandwidth: B
If cluster size is N, then:

$$\text{Bandwidth per cell} = \frac{B}{N}$$

- Larger $N \rightarrow$ each cell gets a smaller fraction of spectrum
- Fewer channels per cell
- Fewer simultaneous users per cell

Key Insight

Increasing N reduces interference, but it also reduces spectral efficiency.

Co-Channel Interference

Let i_o be the number of co-channel interfering cells. Then, the signal-to-interference ratio (S/I or SIR) for a mobile receiver which monitors a forward channel can be expressed as

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^{i_0} I_i} \tag{2.5}$$

where S is the desired signal power from the desired base station and I_i is the interference power caused by the ith interfering co-channel cell base station. If the signal levels of co-channel cells are known, then the S/I ratio for the forward link can be found using equation (2.5).

Co-Channel Interference

Signal-to-interference ratio (approx):

$$\mathrm{SIR} \approx \frac{(D/R)^n}{i_0}$$

Where:

- n = path loss exponent
- i_0 = number of first-tier interferers

Substituting Reuse Distance

Since:

$$D/R = \sqrt{3N}$$

Then:

$$\text{SIR} \approx \frac{(3N)^{n/2}}{i_0}$$

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Example Calculation

Assume:

- $n = 4$
- $i_0 = 6$
- Required SIR = 18 dB

Find minimum N.

Converting to Linear Scale

18 dB:

$$\mathrm{SIR}_{lin} = 10^{18/10} \approx 63$$

We need:

$$\frac{(3N)^2}{6} \geq 63$$

Solving for N

$$(3N)^2 \geq 378$$

$$9N^2 \geq 378$$

$$N^2 \geq 42$$

$$N \geq 6.5$$

Choose $N = 7$

What is a Handoff?

Definition

A handoff (handover) is the process of transferring an ongoing call or data session from one base station to another.

Why needed?

- User mobility
- Limited cell coverage
- Load balancing
- Interference management

Received Signal Strength and Mobility

As a mobile moves away from BS1:

- Path loss increases
- Received power decreases
- Interference from neighboring cell may increase

If no handoff occurs:

- SNR drops
- Call may be terminated

Handoff Threshold Concept

Two important levels:

- Minimum acceptable signal level (call drops below this)
- Handoff threshold (higher than minimum level)

Key idea: Handoff must be initiated BEFORE signal becomes too weak.

Improper vs Proper Handoff

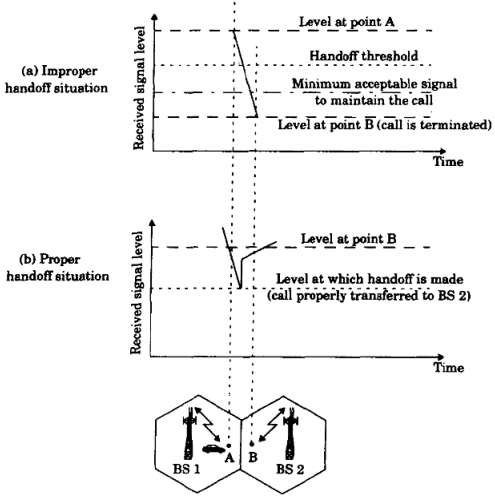

Figure 2.3
Illustration of a handoff scenario at cell boundary.

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Handoff Mechanisms
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Hysteresis and Dwell Time

To prevent rapid switching ("ping-pong effect"):

- Hysteresis margin added
- Time-to-trigger requirement

Handoff occurs only if:

$$P_{BS2} > P_{BS1} + \Delta$$

for a certain time duration.

Hard vs Soft Handoff

Hard Handoff (Break-before-make)

- Used in GSM
- Connection to old BS released before new one established

Soft Handoff (Make-before-break)

- Used in CDMA
- Connected to multiple BS simultaneously

Handoff in LTE and 5G

Modern systems:

- Measurement reports from UE
- X2 / NG interface coordination
- Fast decision algorithms
- Load-aware and interference-aware handovers

Today:

- Mostly hard handover
- But extremely fast (milliseconds)

Sectoring

Replace omni antenna with:

- $3 \times 120°$
- $6 \times 60°$

Effect:

- Reduce i_0
- Improve SIR
- Increase hardware complexity

Cell Splitting

When traffic increases:

- Reduce cell radius
- Reduce transmit power
- Increase number of base stations

Power Scaling in Cell Splitting

If:

$$R \to R/2$$

Then required power scales approximately:

$$P \propto R^n$$

Cell Splitting

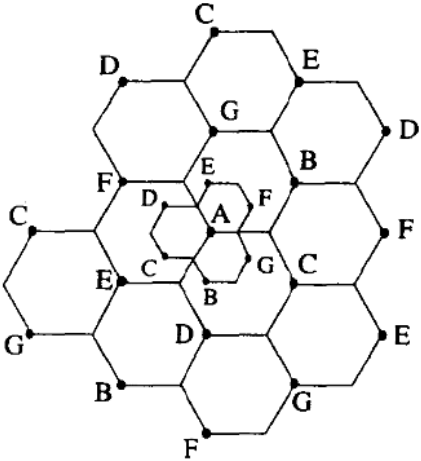


Figure 2.8
Illustration of cell splitting.

Microcells and Picocells

Used in:

- Dense urban areas
- Indoor environments
- High traffic zones

Interference-Limited Systems

Modern systems are:

- Interference-limited
- Not noise-limited

Thermal noise is often negligible compared to interference.

Evolution to LTE and 5G

Old systems:

- Fixed reuse patterns
- Hard frequency planning

Modern systems:

- Reuse 1
- OFDMA
- Power control
- Beamforming
- Interference coordination

Reuse 1 Concept

All cells use full bandwidth.
Interference handled by:

- Scheduling
- Power control
- Adaptive modulation
- MIMO

Key Trade-Off Summary

- Capacity vs Interference
- Coverage vs Power
- Infrastructure cost vs Performance

What You Must Understand

1. Why frequency reuse is necessary
2. How N is calculated
3. How SIR depends on N
4. Why modern systems use reuse 1

MM6

Vit tosaðu um:

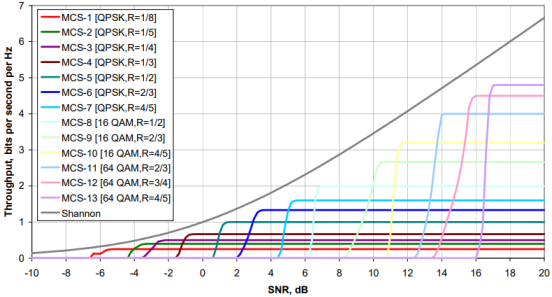
- Avlevering 1
- Fjarskiftiseftirlitið
- MRCC vitjan
- In-class rokniuppgávu
- Internet of Things
  - Licenced og Unlicensed spectrum
  - EC-GSM
  - NB-IoT

Channel Coding

Channel coding adds redundancy to improve reliability.

$$R_c = \frac{k}{n}$$

- k: information bits
- n: transmitted bits
- Lower R_c means more redundancy
- More redundancy improves error correction



Recap: Code Rate and Coding Gain

$$R_c = \frac{k}{n}$$

Coding scheme	Code rate	Coding gain
Uncoded BPSK	1.00	0 dB
Convolutional code	0.5	5 dB
Turbo code	0.33	7 dB
LDPC code	0.83	8 dB
Polar code	0.5	7.5 dB

Coding reduces the required SNR for a given BER, at the cost of extra transmitted bits.

Example: Block Code with Row and Column Parity

16 information bits are arranged in a 4×4 block.

$$\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 1 & 1 & 0 \end{array}$$

Row and column parity bits are computed using XOR.
Example:

$$1 \oplus 0 \oplus 1 \oplus 1 = 1$$

The final row and column provide redundancy that allows error detection.

Linear Algebra View of Block Codes

A block code maps an information vector u to a codeword c

$$\mathrm{c} = \mathrm{u}G$$

where G is the generator matrix.
For a linear block code, valid codewords also satisfy

$$H\mathrm{c}^T = 0$$

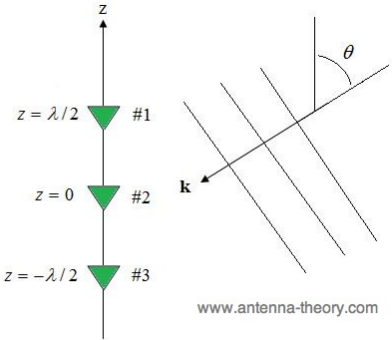
where H is the parity-check matrix.
If the receiver observes r, it computes the syndrome

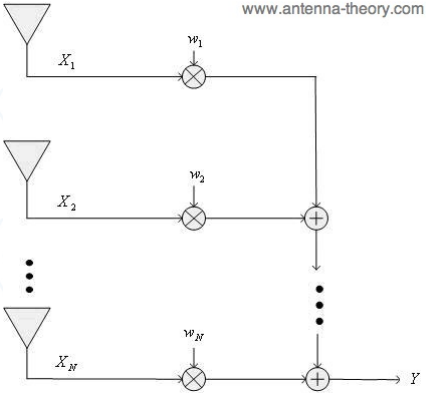
$$\mathrm{s} = H\mathrm{r}^T$$

- $\mathrm{s} = 0$: no error detected
- $\mathrm{s} \neq 0$: an error is detected

Beamforming Principle

- Typical antennas radiate energy in many directions
- Antenna arrays can control the phase of each element
- Signals combine constructively in the desired direction
- This creates a focused beam and increases received power





Recap: Beamforming and Array Gain

- Beamforming uses multiple antennas to direct energy toward a receiver
- Signals from different antennas combine constructively in the desired direction

For an array with N antennas:

$$y(t) = \sum_{n=1}^{N} w_n x_n(t) e^{j\phi_n}$$

If the phases are aligned, the received power increases with N.
The theoretical array gain is

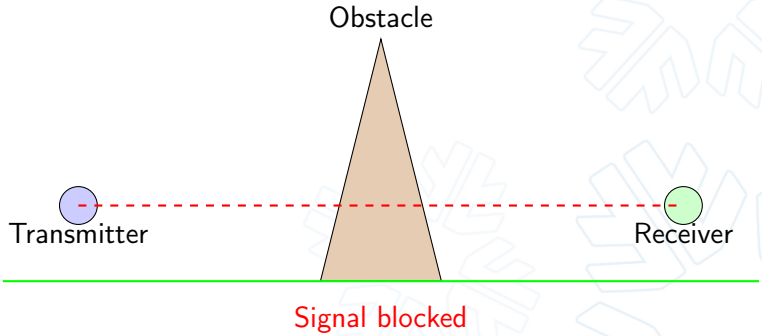
$$G_{\text{array}} = 10 \log_{10}(N) \text{ dB}$$

Examples:

- $N = 2 \rightarrow 3$ dB gain
- $N = 4 \rightarrow 6$ dB gain
- $N = 64 \rightarrow 18$ dB gain

MM1
MM2
MM3
MM4
MM5
MM6
MM7
Coding gain
Beamforming
Relays and Multi-Hop Links
Exam Preparation

When a Direct Link Fails



When a Direct Link Fails

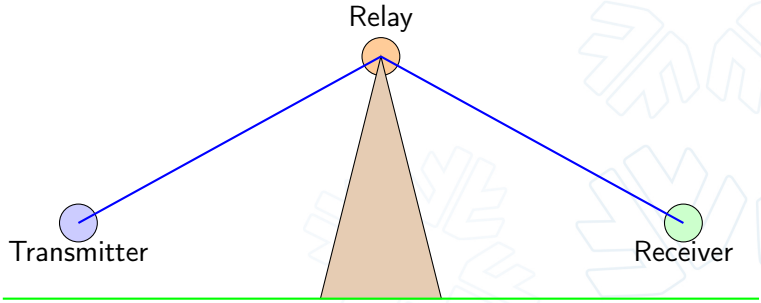
- A wireless link is feasible only if the received power exceeds the required SNR

$$P_r = P_t + G_t + G_r - L_{\text{path}}$$

- Path loss increases rapidly with distance
- Obstacles (walls, buildings) further reduce received power
- Increasing transmit power is often limited
- Increasing antenna gain is not always possible

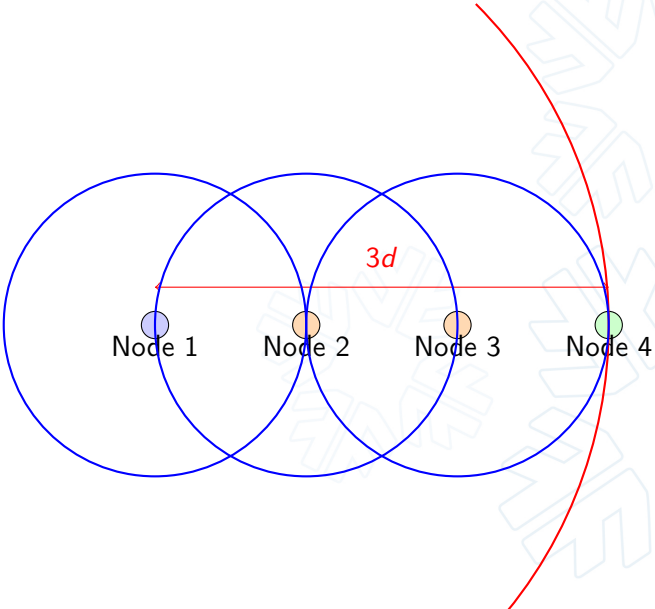
Alternative: forward the message through intermediate nodes

Using a Relay



Two shorter links instead of one blocked link

Intuition: Path Loss and Multi-Hop



Energy Required for Transmission

In wireless communication, the required transmit power depends strongly on distance.

For many wireless channels, path loss can be approximated as

$$L(d) \propto d^n$$

where

- d is the transmitter–receiver distance
- n is the path loss exponent

Typical values:

- Free space: $n \approx 2$
- Indoor environments: $n \approx 3$–4
- Dense urban: $n \approx 4$–5

Required transmit energy increases rapidly with distance.

Energy Comparison: One Hop vs Multi-Hop

Assume transmit energy scales with distance

$$E_{tx} \propto d^n$$

Direct transmission over 300 m:

$$E_{\text{direct}} \propto 300^n$$

Three hops of 100 m:

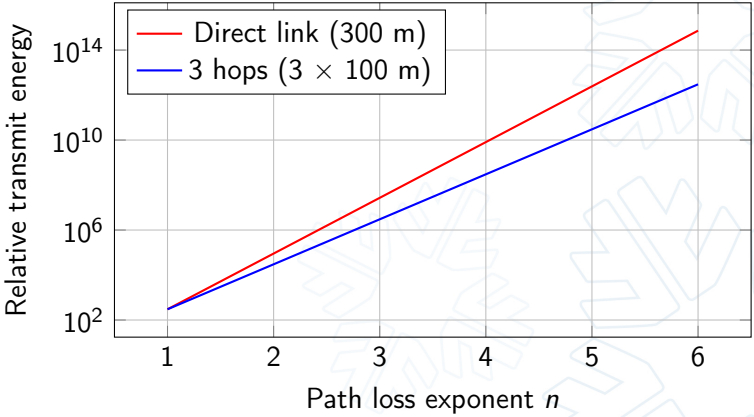
$$E_{\text{multi-hop}} \propto 3 \cdot 100^n$$

Example with $n = 4$:

$$300^4 = 8.1 \times 10^9$$

$$3 \cdot 100^4 = 3 \times 10^8$$

Energy vs Path Loss Exponent



Relaying Has a Cost

Multi-hop communication reduces transmission distance.
However, each additional hop requires:

- another transmission
- another reception
- additional delay

Energy consumed for one forwarded message:

$$E_{\text{hop}} = E_{\text{tx}} + E_{\text{rx}}$$

For k hops:

$$E_{\text{total}} = \sum_{i=1}^{k} E_{\text{tx}}(d_i) + (k-1)E_{\text{rx}}$$

Which Nodes Should Relay?

In a network with many sensors and possible relay nodes:

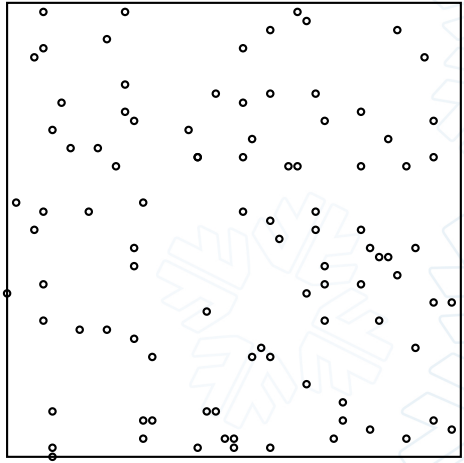
- Some nodes may have better channel conditions
- Some nodes may have more available energy
- Some nodes may be closer to the gateway

Possible routing strategies:

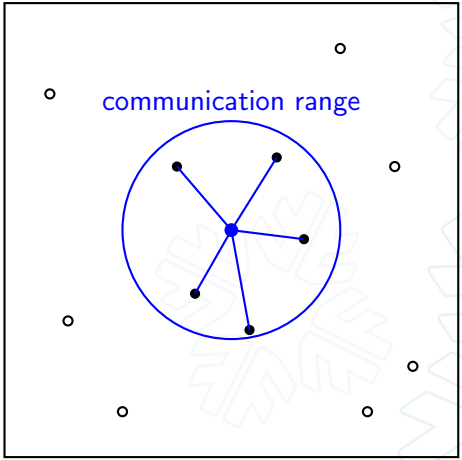
- Minimize number of hops
- Minimize transmission energy
- Prefer mains-powered relay nodes
- Avoid overloading battery-powered nodes

Goal: find the lowest energy path to the gateway

A Simple Wireless Network Model

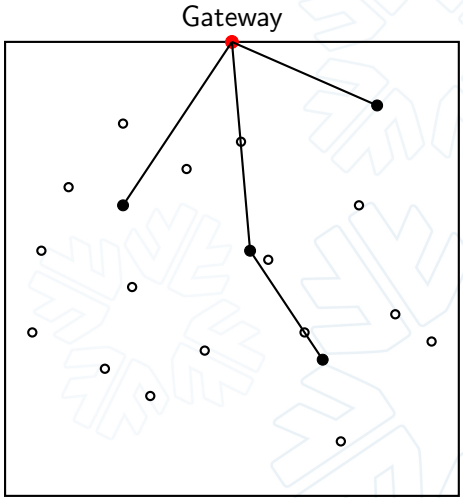


Communication Range and Neighbor Links



Gateway and Relay Nodes

Open: sensors
Filled: relay nodes



Common Multi-Hop Network Topologies

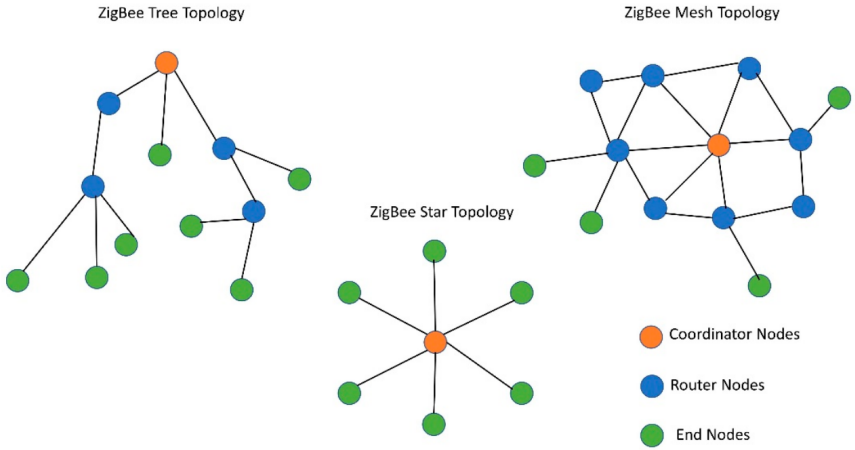


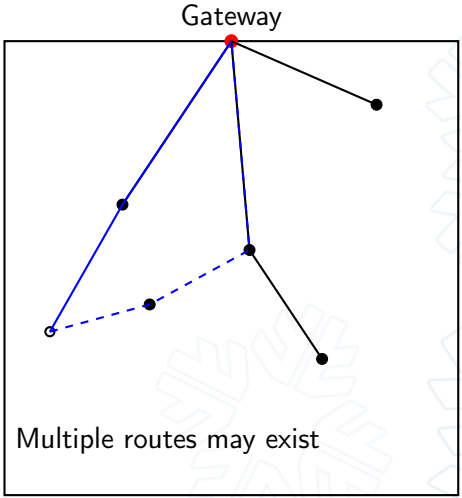
Figure 2. Different topologies of the Zigbee standard.

Topology Comparison

Property	Star	Tree	Mesh
Implementation complexity	Low	Medium	High
Number of relays	None	Some	Many
Robustness to link failure	Low	Low–Medium	High
Energy per forwarded message	Low	Medium	High
Scalability	Low	Medium	High

- Star is simple, but all nodes depend on one central link
- Tree extends coverage, but failures can disconnect branches
- Mesh is more robust, but forwarding creates extra traffic and energy cost

Routing Choices



- A message may reach the gateway through several possible paths
- Different routes have different energy costs
- The goal is to select routes that minimize total transmission energy

Feasible Wireless Links

Not every pair of nodes can communicate directly.
A link between nodes i and j is feasible only if

$$P_r = P_t + G_t + G_r - L_{ij}$$

satisfies the receiver requirement

$$P_r - (N + I) \geq \gamma_{\min}$$

where

- L_{ij} is the path loss between nodes
- N is the noise power
- I represents interference
- $\gamma_{\min}$ is the required SINR

Link Quality Matrix

For each pair of nodes i, j, compute the received power

$$P_{r,ij} = P_t + G_t + G_r - L_{ij}$$

Example matrix:

$$P_r = \begin{bmatrix} - & -82 & -110 & -95 \\ -82 & - & -88 & -120 \\ -110 & -88 & - & -90 \\ -95 & -120 & -90 & - \end{bmatrix} \text{dBm}$$

Remove links that do not satisfy the SINR requirement.

$$P_r^* = \begin{bmatrix} - & -82 & & -95 \\ -82 & - & -88 & \\ & -88 & - & -90 \\ -95 & & -90 & - \end{bmatrix}$$

Routing is then performed only on the remaining links.

Energy Cost Matrix

Each feasible link has an associated transmission cost.
Assume transmit energy depends on distance

$$E_{ij} = k\, d_{ij}^{n}$$

Example energy matrix:

$$E = \begin{bmatrix} - & 2.1 & & 3.4 \\ 2.1 & - & 1.8 & \\ & 1.8 & - & 2.5 \\ 3.4 & & 2.5 & - \end{bmatrix}$$

Routing algorithms select the path that minimizes the total transmission energy to the gateway.

Minimum-Energy Routing

Let E_{ij} denote the transmission energy required for link $i \to j$.
A route from node s to the gateway g consists of a sequence of hops.
The total cost of a route is

$$E_{\text{path}} = \sum_{(i,j)\in\text{path}} E_{ij}$$

The routing problem becomes

$$\min_{\text{path}} \sum_{(i,j)\in\text{path}} E_{ij}$$

subject to the constraint that all links satisfy the SINR requirement.

Bandwidth Increase With Fixed P_t

If P_t and channel losses are fixed:

- increasing B increases noise power N
- so SNR decreases
- capacity has two competing effects:

$$C = B \log_2(1 + \mathrm{SNR})$$

Conclusion (typical exam explanation): increasing B does not *always* increase rate if SNR collapses.

Modulation Bits/Symbol

- BPSK: $M = 2 \Rightarrow m = \log_2 M = 1$ bit/symbol
- QPSK: $M = 4 \Rightarrow m = 2$ bits/symbol
- 16-QAM: $M = 16 \Rightarrow m = 4$ bits/symbol
- 64-QAM: $M = 64 \Rightarrow m = 6$ bits/symbol

Typical exam statement: higher-order modulation increases spectral efficiency but requires higher SNR (less noise robust).

Beamforming Antennas for SNR Target

If a system has $\mathrm{SNR} = 0\,\mathrm{dB}$ but requires $10\,\mathrm{dB}$, then we need about $10\,\mathrm{dB}$ additional link margin.

Ideal array gain:

$$G_{\text{array}} \approx 10 \log_{10}(N)\ \mathrm{dB}$$

Solve $10 \log_{10}(N) = 10 \Rightarrow N = 10$.

In practice you would choose the next implementable array size (e.g., 16).